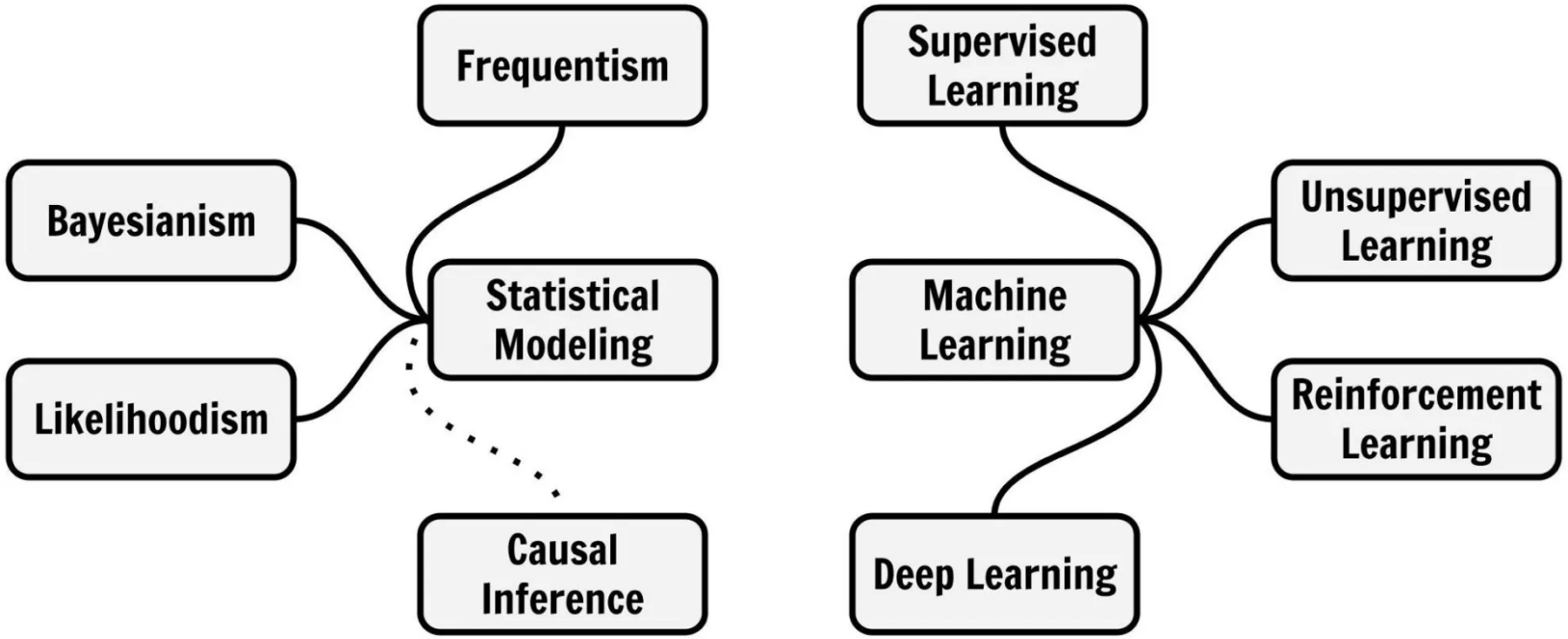
What is machine learning?

And why is it useful (for science)?

Modeling mindsets:



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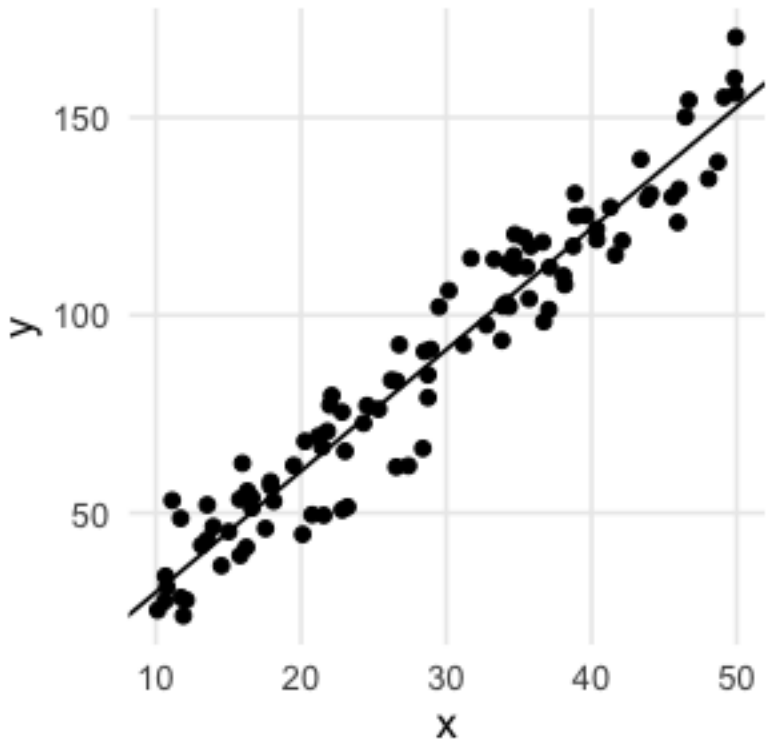
The same “model” can fit into multiple mindsets

- A frequentist can use a linear regression to test a hypothesis
- A Bayesian can measure the posterior distribution of a linear coefficient
- A machine learner can use a linear regression to predict unobserved data



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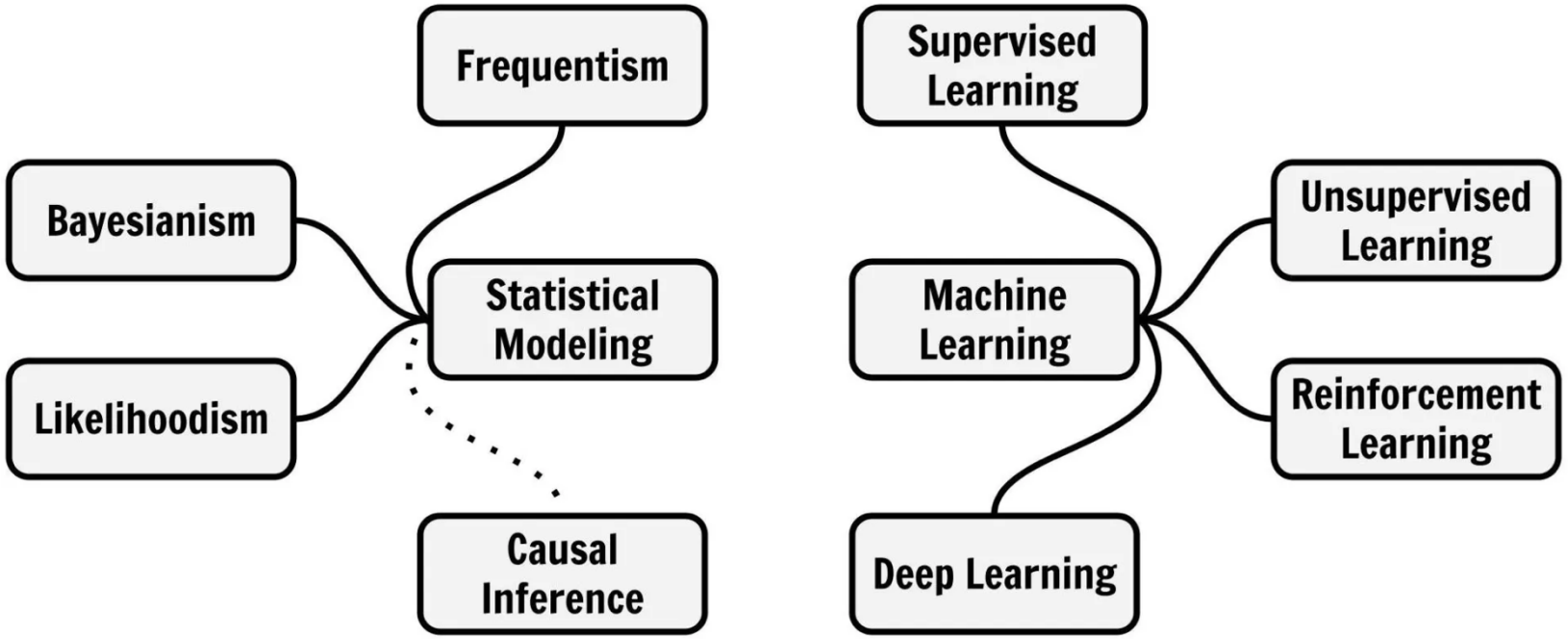

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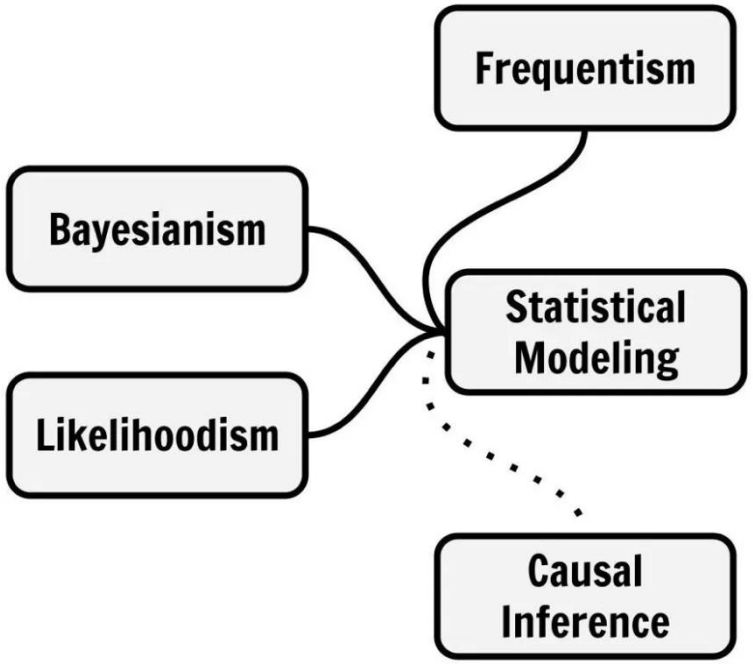

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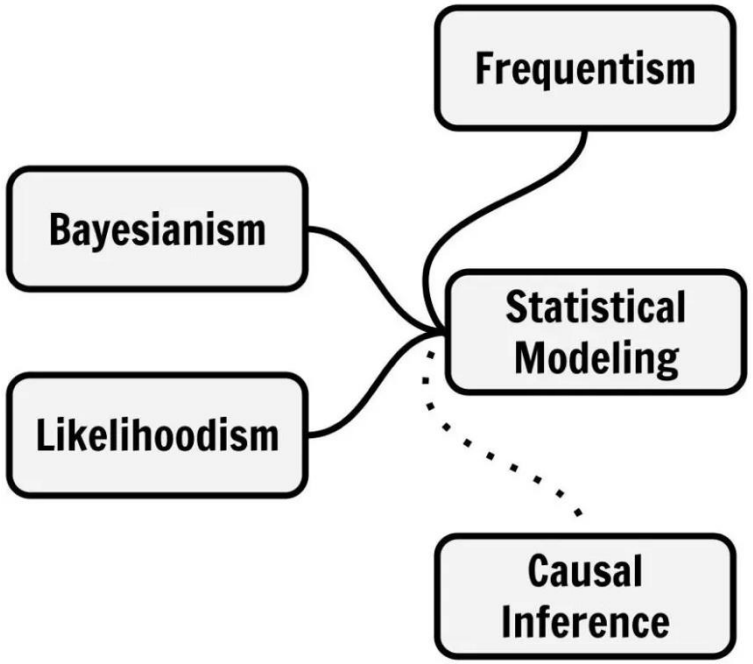
Modeling mindsets:

X → NATURE → Y



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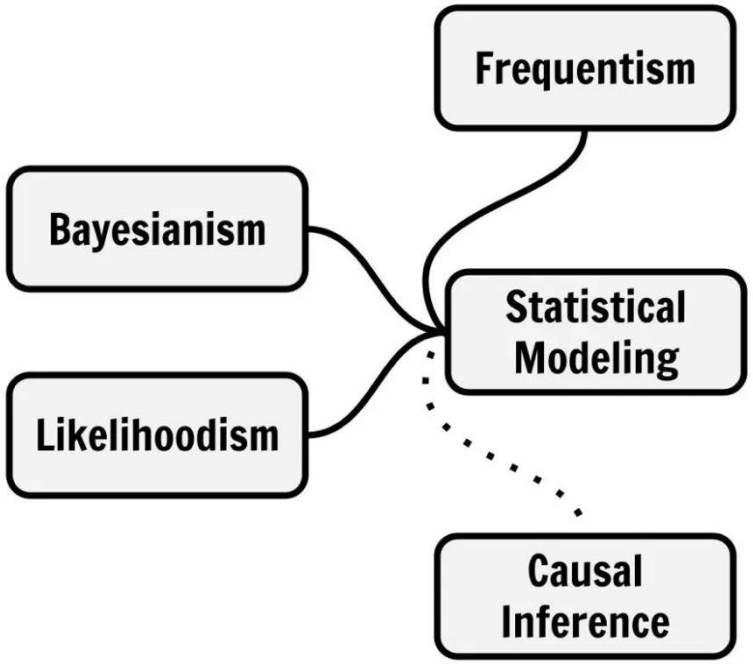
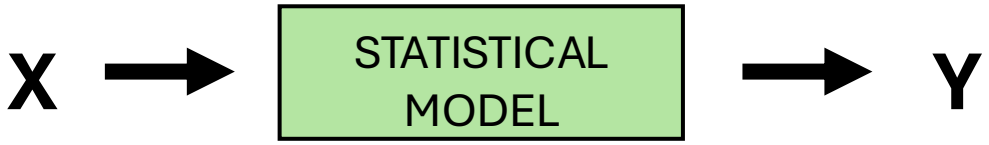


Statistical model: approximate how nature gets from X to Y by mimicking the *data generating process*



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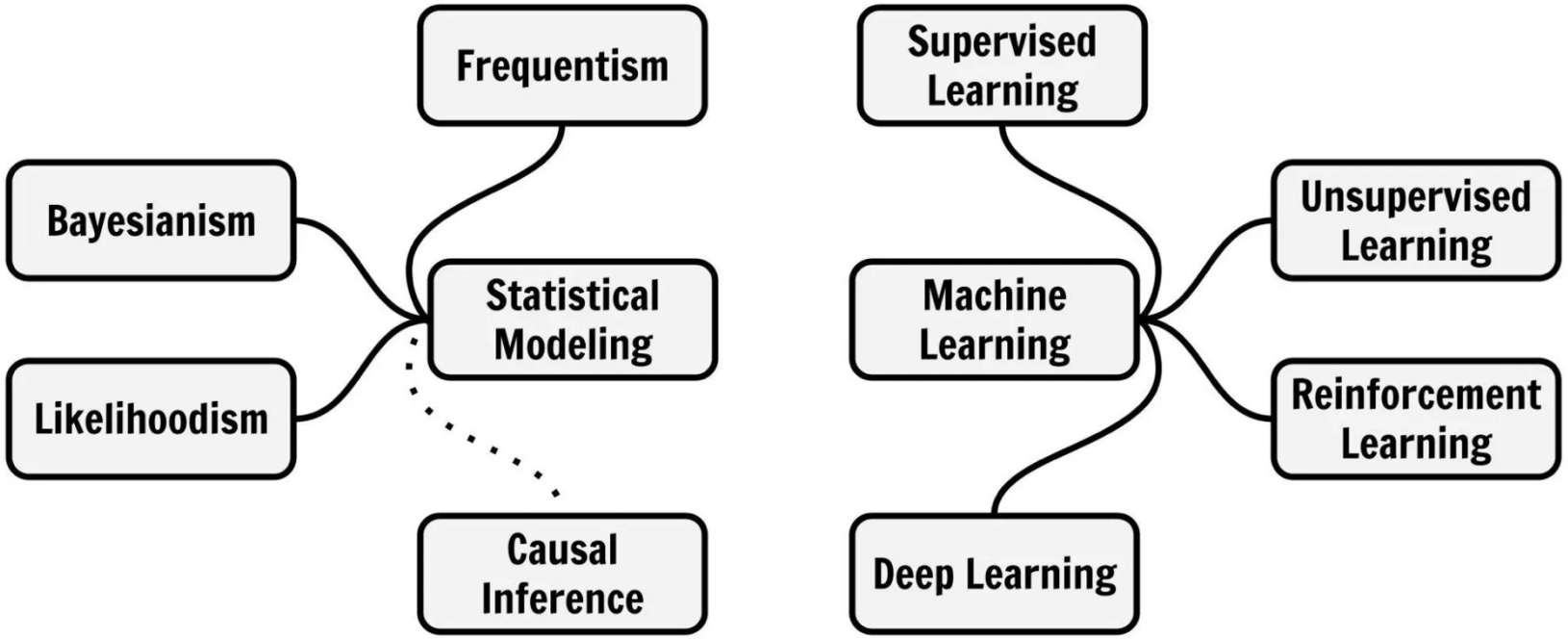

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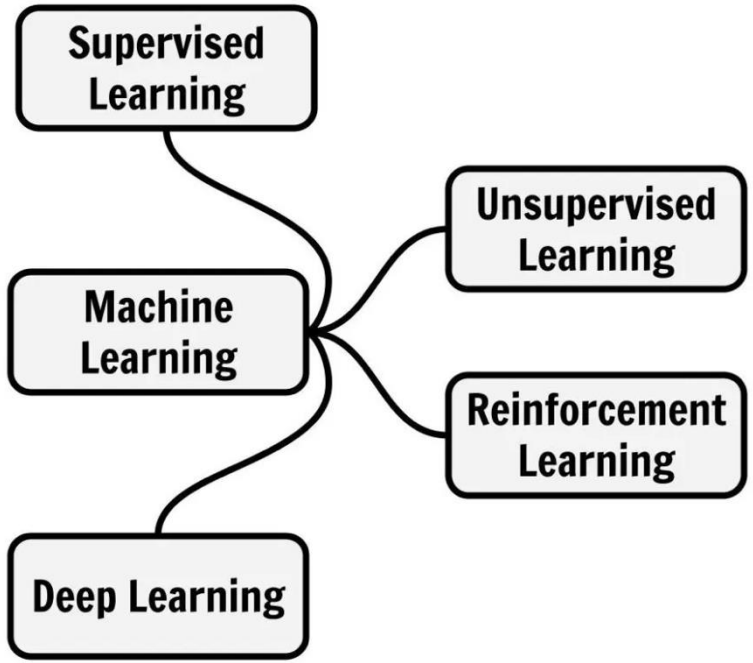

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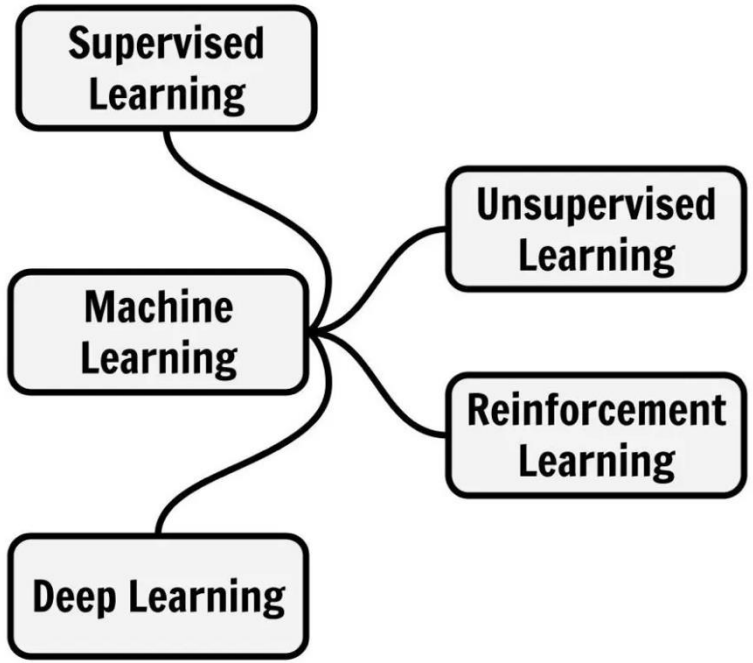
Modeling mindsets:

Machine learning: use algorithms to solve a task



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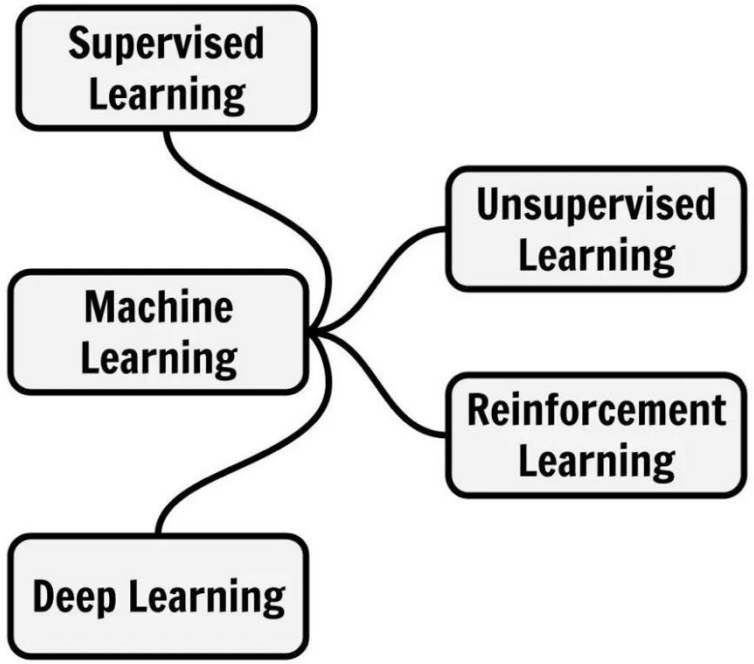

Make 6 groups based on these data

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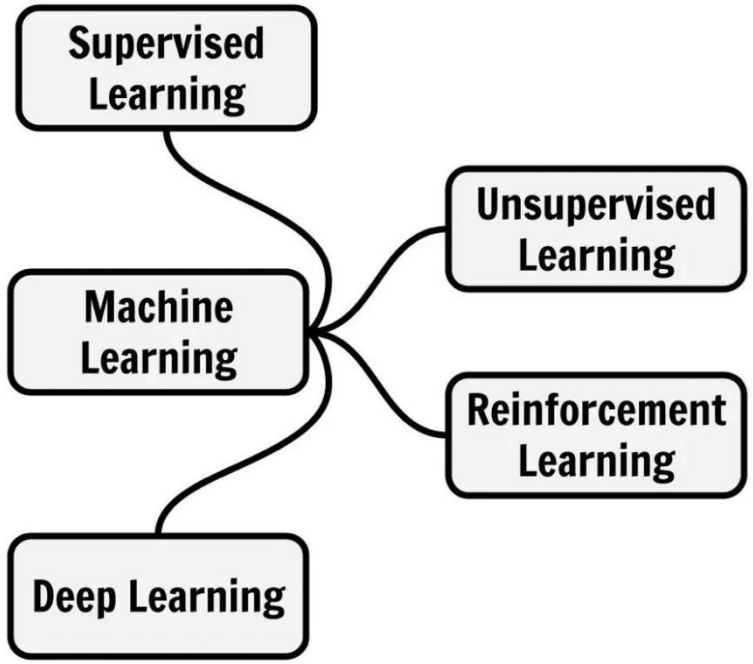

Make 6 groups based on these data

Get better at chess

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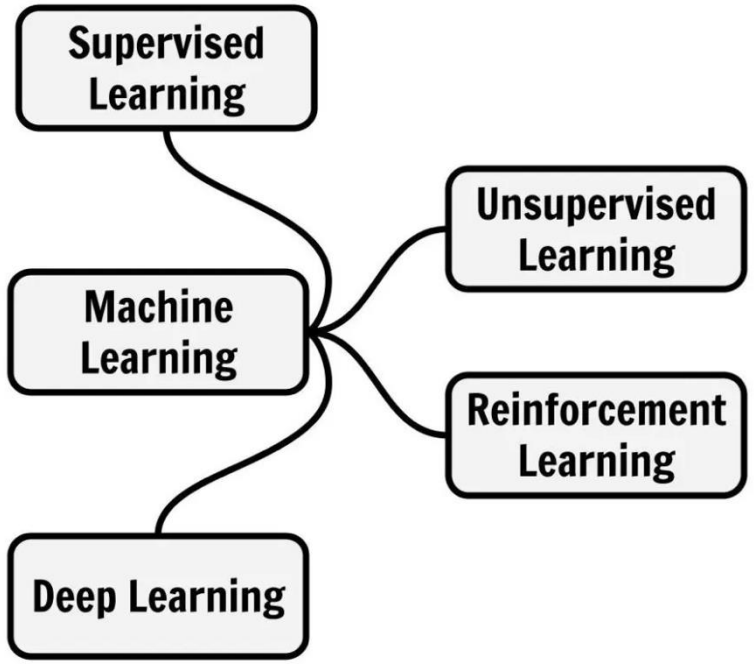

Make 6 groups based on these data

Get better at chess

Identify things in images

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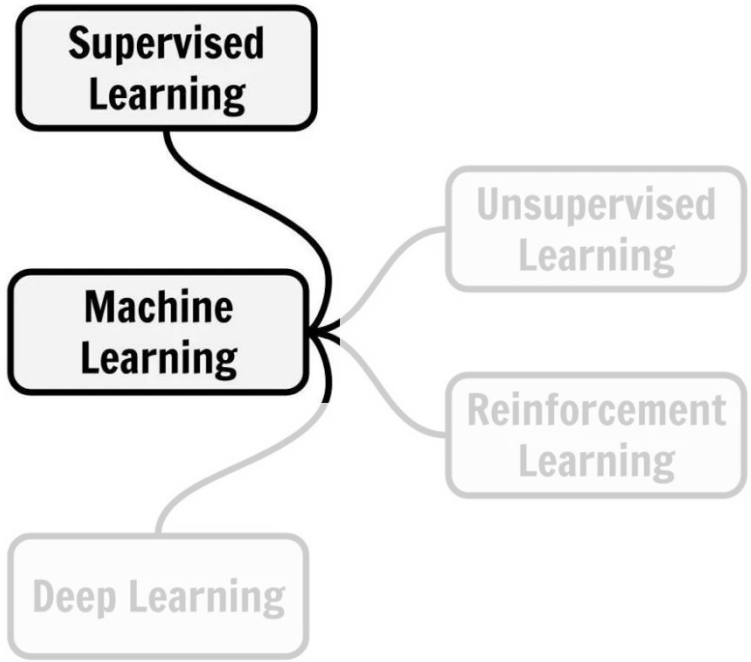

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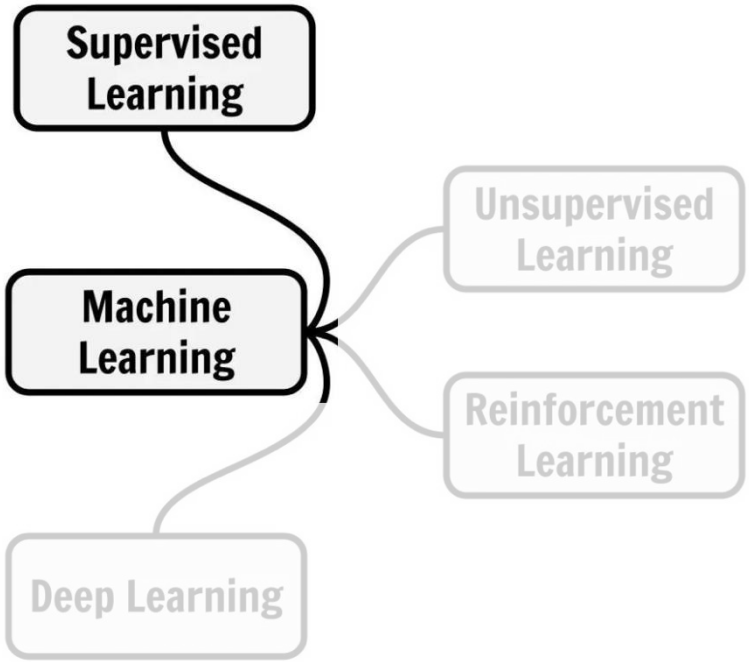
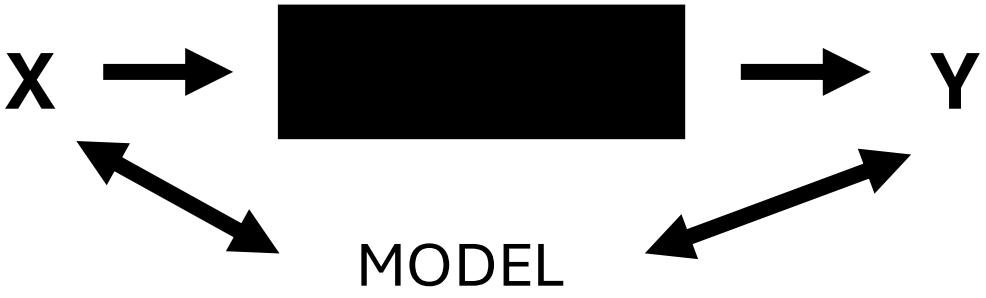
Modeling mindsets:

Machine learning: use algorithms to solve a task

Supervised machine learning: use algorithms to *mimic the outputs* of nature





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In science, we often have multiple aims

- Observe the world
- Generate hypotheses
- Test a hypothesis
- Estimate a value
- Predict something about the world

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Supervised machine learning can help!

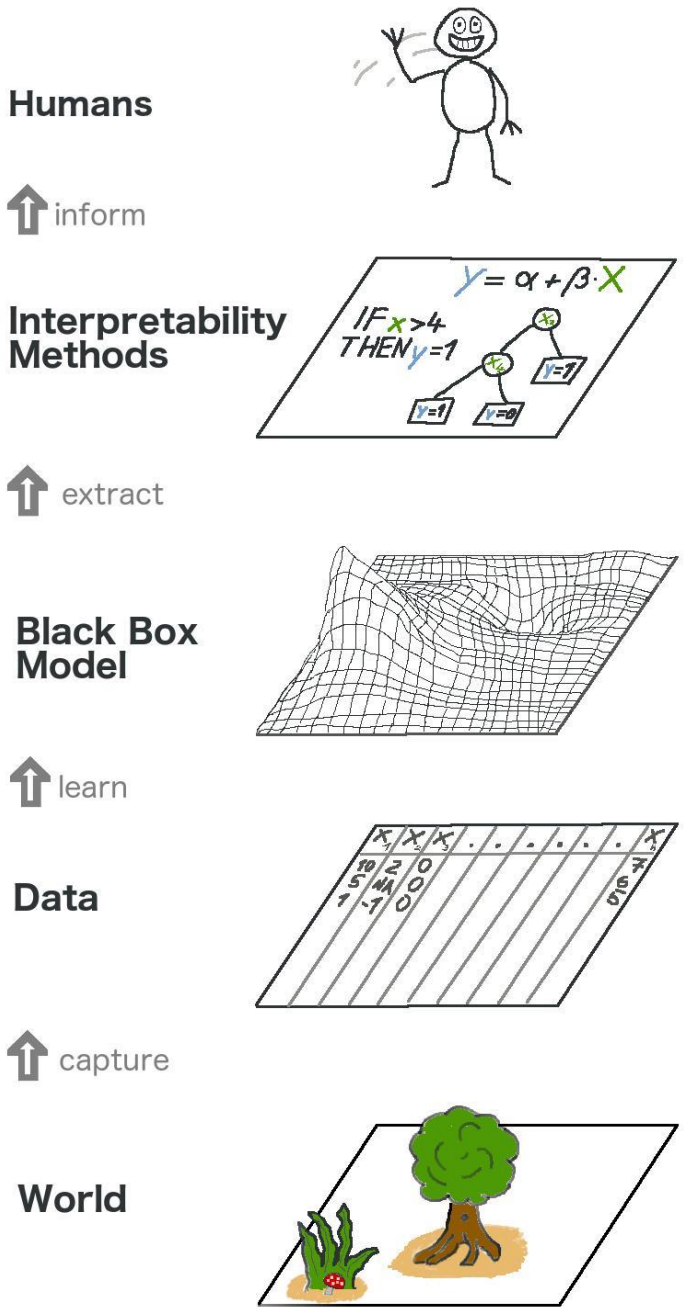
- **Observe**: Classify trail camera images to identify animals at scale
- **Generate:** Identify patterns and suggest important variables for future experiments
- **Predict:** Forecast the future distribution or abundance of a plant

But we can't understand black box models?

Well, sort of…

- Many algorithms used to make predictions are very complex and difficult (impossible?) to understand

- But post-hoc approaches can identify how the model is using variables to make predictions



FREE BOOK: https://christophm.github.io/interpretable-ml-book

Some terminology differences…

Statistical modeling	Machine learning
Fit a model	Train a model
Dependent variable; Response variable	Response variable; Target variable
Independent variable; Predictor variable	Feature variable; Predictor variable
Examine the coefficients	“Interpret” or “Explain” the model

Today, we'll focus on supervised machine learning and tree-based algorithms

We'll cover:

- Model evaluation and cross-validation approaches
- Handling correlated variables
- Class-balancing
- Model tuning
- Model interpretation
- Classification and regression trees (CART)
- Random forests / bagging
- Boosted regression trees

We won't cover:

- K-means clustering
- Dimensionality reduction / PCA
- Neural networks / deep learning
- Natural language processing
- Image processing
- Bayesian Additive Regression Trees
- All the possible interpretation methods